

AWS VPC Peering: İki VPC'yi Özel Hatla Birleştir!

AWS üzerinde büyüyen sistemlerde işler genellikle tek bir VPC ile sınırlı kalmaz.

Uygulamalar farklı bölgelerde çalışabilir, bazı bileşenler public erişime açıkken bazıları tamamen private kalabilir. İşte tam bu noktada **VPC Peering**, iki VPC'yi birbirine özel bir hat üzerinden bağlamamızı sağlar.

Neden VPC Peering?

- **Düşük gecikme:** Trafik AWS backbone üzerinden akar, internetten dolaşmaz.
- **Güvenlik:** Bağlantı tamamen private IP'lerle çalışır.
- **Basitlik:** VPN ya da Transit Gateway kurmaya gerek kalmadan iki VPC'yi konuşturabilirsiniz.

Ama unutmayın:

⚠ VPC Peering'de **transit routing yoktur**. Yani A→B, B→C bağlıysa A doğrudan C'ye gidemez.

⚠ CIDR bloklarının çakışmaması gerekir.

⚠ Route tabloları **her iki tarafta da** güncellenmelidir.

🎯 Bu yazıda ne öğreneceksiniz?

Bu lab çalışmasında:

- Aynı bölgede (örneğin **us-east-1 N. Virginia**) iki VPC kuracağız.
 - **VPC-A:** Public subnet içeriyor, bastion EC2 çalışıyor.
 - **VPC-B:** Private subnet içeriyor, public erişim yok.
- Bu iki VPC arasında **peering connection** oluşturacağız.
- Sonrasında bastion EC2 üzerinden private EC2'ya **özel IP ile SSH bağlantısı** kuracağız.
- Lab sonunda kaynakları silip maliyet sürprizlerini önleyeceğiz.

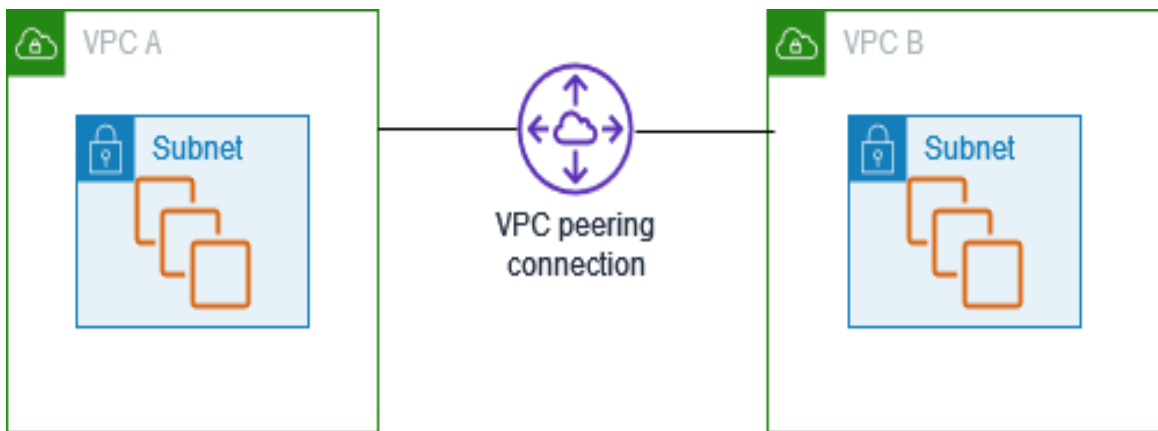
🌍 Senaryo ve Mimari

- **VPC-A (Public)**
 - **CIDR: 10.0.0.0/16**
 - **Public Subnet: 10.0.1.0/24**
 - İçinde çalışan EC2: VPC-A_EC2
 - Public IP var
 - Bastion görevi görüyor
 - SSH + HTTP erişimi açık
- **VPC-B (Private)**
 - **CIDR: 20.0.0.0/16**
 - **Private Subnet: 20.0.0.0/24**
 - İçinde çalışan EC2: VPC-B_EC2
 - Public IP yok
 - Sadece içeriden SSH erişimi

👉 **Hedef:** VPC-A'daki bastion EC2 üzerinden, VPC-B'deki private EC2'ya özel IP ile SSH bağlantısı kurmak.

Mimariyi Basitçe Hayal Edelim

- VPC-A → Public Subnet → Bastion EC2 (internet erişimli)
- VPC-B → Private Subnet → Private EC2 (yalnızca internal IP)
- Aralarında **VPC Peering Connection** var.
- Route tabloları karşılıklı güncellenmiş durumda.



VPC-A (public subnet) ile VPC-B (private subnet) arasında kuracağımız VPC peering bağlantısının mimarisi

Bölüm 1 — VPC-A (Public) Kurulumu

İlk olarak internet erişimi olan **VPC-A**'yı oluşturacağız. Bu VPC içinde bir **public subnet** kuracak, internet erişimi için **Internet Gateway** ekleyecek ve en sonunda bir **bastion EC2** çalıştıracacağız. Bu bastion sunucusu üzerinden diğer VPC'ye bağlanacağız.

1. VPC Oluşturma

VPC → Your VPC → Create VPC

- AWS Console'a giriş yapın.
- Bölge olarak **N. Virginia (us-east-1)** seçin.
- Menüden **VPC → Your VPCs → Create VPC** yolunu izleyin.
- **Seçenek:** *VPC only*
- **Name:** **VPC-A**
- **IPv4 CIDR:** **10.0.0.0/16**
- Diğer ayarları varsayılan bırakıp **Create VPC** butonuna basın.

☰ VPC > Your VPCs > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional Info

Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info

☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Tenancy Info

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	VPC-A	Remove tag
Add tag		

You can add 49 more tags

Listeden **VPC-A**'yı seçin → **Actions** → **Edit VPC settings**.

- **Enable DNS resolution** ve **Enable DNS hostnames** kutularını işaretleyip **Save** edin

Menu > VPC > Your VPCs > vpc-0203c03be6de16894 > Edit VPC settings

Edit VPC settings Info

VPC details
VPC ID
vpc-0203c03be6de16894
Name
VPC-A

DHCP settings
DHCP option set Info
dopt-0f244fe1db761346b

DNS settings
☒ Enable DNS resolution Info
☒ Enable DNS hostnames Info

Network Address Usage metrics settings
☐ Enable Network Address Usage metrics Info

Cancel **Save**

vpc-0203c03be6de16894

You have successfully modified the settings for vpc-0203c03be6de16894 / VPC-A.

vpc-0203c03be6de16894 / VPC-A Actions

Details Info
VPC ID
vpc-0203c03be6de16894
DNS resolution
Enabled
Main network ACL
acl-078d507e26f2b110
IPv6 CIDR (Network border group)
-
State
Available
Tenancy
default
Default VPC
No
Network Address Usage metrics
Disabled
Block Public Access
Off
DHCP option set
dopt-0f244fe1db761346b
IPv4 CIDR
10.0.0.0/16
Route 53 Resolver DNS Firewall rule groups
Failed to load rule groups
DNS hostnames
Enabled
Main route table
rtb-0ce8bd5d43fe4aa2a
IPv6 pool
-
Owner ID
088711677111

Resource map Info | CIDRs | Flow logs | Tags | Integrations

Resource map Info Show all details
VPC
Your AWS virtual network
VPC-A
Subnets (0)
Subnets within this VPC
Route tables (1)
Route network traffic to resources
rtb-0ce8bd5d43fe4aa2a
Network Connections (0)
Connections to other networks

✓ Artık **VPC-A** hazır. DNS resolution ve hostnames ayarları aktif.

2. Public Subnet Oluşturma

- **Subnets** → **Create subnet** yoluna gidin.

Menu > VPC > Subnets

Subnets (6) Info

Last updated 8 minutes ago Actions **Create subnet**

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...
☐	-	subnet-0bcf36eb14d543e7d	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-032c97cae236dc6af	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-09a87397ac2610fd8	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-0b16ab23e115d25d4	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-05b4c0af0d1763c3f	Available	vpc-09f9c7f03444ddf5b	Off
☐	-	subnet-009b05dd22e8efdc8	Available	vpc-09f9c7f03444ddf5b	Off

VPC dashboard <
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets

- VPC ID: **VPC-A**
- Subnet name: **Public-Subnet-A**
- Availability Zone: *No Preference*
- IPv4 subnet CIDR block : **10.0.1.0/24**

VPC > Subnets > Create subnet

VPC
VPC ID
Create subnets in this VPC.
vpc-0021fa599947cef4d (VPC-A)

Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-Subnet-A
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

▼ **Tags - optional**

Key
Q Name X Value - optional
Q Public-Subnet-A X Remove

Add new tag
You can add 49 more tags.
Remove

Add new subnet

Cancel Create subnet

- AZ tercihini varsayılan bırakıp **Create subnet** butonuna basın.

✓ You have successfully created 1 subnet: subnet-074bfe375c52ffd5b

Subnets (1) [Info](#) Last updated less than a minute ago [Actions](#)

Find subnets by attribute or tag

Subnet ID : subnet-074bfe375c52ffd5b X Clear filters

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	Public-Subnet-A	subnet-074bfe375c52ffd5b	Available	vpc-0021fa599947cef4d VPC-A	Off	10.0.1.0/24

3. Internet Gateway Ekleme

Public subnetteki EC2'nun internete çıkabilmesi için **Internet Gateway (IGW)** gerekli.

- **Internet Gateways** → **Create internet gateway**
- **Name:** **IGW-A**

[Internet gateways](#) > Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

IGW-A

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Q Name	Q IGW-A	Remove

Add new tag

You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

- Oluşturduktan sonra **Attach to VPC** → **VPC-A** seçin.

The following internet gateway was created: igw-0943899a3ccab624b - IGW-A. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#) X

igw-0943899a3ccab624b / IGW-A

[Actions](#)

Details Info

Internet gateway ID igw-0943899a3ccab624b	State Detached	VPC ID -	Owner 088711677111
---	--------------------------	--------------------	------------------------------

Tags

Search tags

Key	Value
Name	IGW-A

Manage tags

< 1 > ⚙

[Internet gateways](#) > Attach to VPC (igw-0943899a3ccab624b)

The following internet gateway was created: igw-0943899a3ccab624b - IGW-A. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#) X

Attach to VPC (igw-0943899a3ccab624b) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC

Q vpc-0203c03be6de16894 X

AWS Command Line Interface command

[Cancel](#) [Attach internet gateway](#)

4. Public Route Table Hazırlama

Public subnet'in trafiğini IGW'ye yönetecek route table'ı oluşturacağız.

- **Route tables** → **Create route table**
- **Name:** PublicRT-A
- **VPC:** VPC-A

VPC > Route tables > Create route table

Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PublicRT-A

VPC
The VPC to use for this route table.

vpc-0203c03be6de16894 (VPC-A)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q PublicRT-A X Remove

Add new tag
You can add 49 more tags.

Cancel Create route table

Route table'ı oluşturduktan sonra **Subnet associations** → **Public-Subnet-A** seçip kaydedin.

VPC > Route tables > rtb-05ff219286e20c9cd > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Q Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public-Subnet-A	subnet-0a007b82d33739be7	10.0.1.0/24	-	Main [rtb-0ce8f...

Selected subnets

subnet-0a007b82d33739be7 / Public-Subnet-A X

Cancel Save associations

Edit routes → **Add route** diyerek şunu ekleyin:

Destination: 0.0.0.0/0 **Target:** IGW-A

VPC > Route tables > rtb-05ff219286e20c9cd > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway		No	CreateRoute

Add route

Cancel Preview Save changes

5. Bastion EC2 Oluşturma

Artık Public subnet içinde bir bastion host sunucusu açalım.

- Bölge: **us-east-1** (N. Virginia).
- **Services** → **EC2** → **Instances** → **Launch instances**.
- **Name and tags**:
 - **Name**: **VPC-A_EC2**
- **Application and OS Images (AMI)**:
 - **Quick Start** → **Amazon Linux 2023 (kernel-6.1)**

Name and tags [Info](#)
Name
 [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs | **Quick Start**

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE

Debian
debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-00ca32bbc84273381 (64-bit (x86), uefi-preferred) / ami-0aa7db6294d00216f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

- **Instance type**: **t2.micro**

▼ Instance type [Info](#) | [Get advice](#)
Instance type

t2.micro **Free tier eligible** ▼
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations
[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- **Key Pair (login):**
 - **Create new key pair** → **Key pair name:** `ec2_ssh_key`
 - **Type:** RSA, **File format:** `.pem` → **Create** ve seç.

Create key pair [X]

Key pair name
Key pairs allow you to connect to your instance securely.

`ec2_ssh_key`

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

Warning: When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel **Create key pair**

Key dosyasını bilgisayarına kaydet

- **Network settings** → **Edit:**
 - **VPC:** `VPC-A`
 - **Subnet:** varsayılan (Public_subnet_VPC-A seçili gelmeli)
 - **Auto-assign public IP:** `Enable`
 - **Firewall (SG): Create a new security group**
 - **Security group name:** `Public_EC2_SG`
 - **Description:** `Security group for public EC2`

■ Rules:

1. **SSH** → **Source:** Anywhere (0.0.0.0/0)
2. **HTTP** → **Source:** Anywhere

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0021fa599947cef4d (VPC-A)
10.0.0.0/16



Subnet [Info](#)

subnet-074bfe375c52ffd5b Public-Subnet-A
VPC: vpc-0021fa599947cef4d Owner: 538594437814
Availability Zone: us-east-1c (use1-az2) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24



[Create new subnet](#)

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

Public_EC2_SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-:/()#,@[]+=&:{}!\$*

Description - required [Info](#)

Security group for public EC2

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0)

[Remove](#)

Type [Info](#)

ssh

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere

Source [Info](#)

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

[Remove](#)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Custom

Source [Info](#)

0.0.0.0/0

Description - optional [Info](#)

e.g. SSH for admin desktop

- **Advanced details:**
 - **User Data Script (opsiyonel, web sayfası göstermek için):**

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul: VPC Peering Lab</h1><html>" >
/var/www/html/index.html
```

User data - *optional* | [Info](#)

Upload a file with your user data or enter it in the field.

[↑ Choose file](#)

```
#!/bin/bash
sudo dnf update -y
sudo dnf install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<html><h1> Welcome to Tech istanbul: VPC Peering Lab</h1><html>" >
/var/www/html/index.html
```

Launch instances butonuna tıkladığınızda, birkaç dakika içinde bastion sunucunuz hazır olacaktır. Public IP'yi tarayıcıya yapıştırarak test sayfasını görebilirsiniz.

Instances (1) Info					Last updated less than a minute ago	Refresh	Connect	Close
Find Instance by attribute or tag (case-sensitive)								
☐	Name ✎	Instance ID	Instance state	Instance type				
☐	VPC-A_EC2	i-0267704e3cd7d281a	Running 🔍 🔍	t2.micro				

- EC2'nun IPv4 Public IP adresini not et.

i-0267704e3cd7d281a (VPC-A_EC2)



Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

i-0267704e3cd7d281a

IPv6 address

–

Hostname type

IP name: ip-10-0-1-168.ec2.internal

Answer private resource DNS name

–

Auto-assigned IP address

3.89.19.3 [Public IP]

Public IPv4 address

3.89.19.3 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-168.ec2.internal

Instance type

[t2.micro](#)

VPC ID

vpc-0021fa599947cef4d (VPC-A)

Private IPv4 addresses

10.0.1.168

Public DNS

ec2-3-89-19-3.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses

–

AWS Compute Optimizer finding

User: arn:aws:iam::538594437814:user/cloud_user is not authorized to perform: compute-optimizer:GetEnrollment

- Bu IP'yi tarayıcıya yapıştırdığında karşılama sayfasını görmelisin.



Welcome to Tech istanbul: VPC Peering Lab

Bölüm 2: VPC-B (Private) Kurulumu

Şimdi ikinci VPC'mizi, yani **VPC-B**'yi oluşturacağız. Bu VPC'de yalnızca **private subnet** olacak ve EC2 instance'ın **public IP'si olmayacak**. Yani sadece içeriden, bastion üzerinden erişilebilir olacak.

1. VPC Oluşturma

- **VPC** → **Your VPCs** → **Create VPC** yolunu izleyin.
- **Seçenek:** *VPC only*
- **Name:** **VPC-B**
- **IPv4 CIDR :** **20.0.0.0/16**
- **Create VPC** butonuna basın.

☰ [VPC](#) > [Your VPCs](#) > Create VPC

Create VPC Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

You can add 49 more tags

[Cancel](#) [Preview code](#)

✅ Artık ikinci VPC hazır.

- Listeden **VPC-B** → **Actions** → **Edit VPC settings** → **Enable DNS resolution** ve **Enable DNS hostnames** işaretle → **Save**.

☰ [VPC](#) > [Your VPCs](#) > [vpc-097810e84793630a3](#) > Edit VPC settings

ⓘ ⌚ 🖨

Edit VPC settings [Info](#)

VPC details

VPC ID
vpc-097810e84793630a3

Name
VPC-B

DHCP settings

DHCP option set [Info](#)
dopt-063cb82d1929ed2f9

DNS settings

☒ Enable DNS resolution [Info](#)

☒ Enable DNS hostnames [Info](#)

Network Address Usage metrics settings

☐ Enable Network Address Usage metrics [Info](#)

[Cancel](#) [Save](#)

2. Private Subnet Oluşturma

Bu subnet internete çıkmayacak, yalnızca iç iletişim için kullanılacak.

- Subnets** → **Create Subnet** seçin.
- VPC ID:** VPC-B
- Subnet name:** Private_subnet_VPC-B
- Availability Zone:** No Preference
- IPv4 CIDR block:** 20.0.0.0/24
- Create subnet.**

☰

VPC > Subnets > Create subnet

🔍 ⌂ 🗨

Create subnet Info

VPC

VPC ID

Create subnets in this VPC.

vpc-02c87f1940a921715 (VPC-B)

Associated VPC CIDRs

IPv4 CIDRs

20.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name

Create a tag with a key of 'Name' and a value that you specify.

Private_subnet_VPC-B

The name can be up to 256 characters long.

Availability Zone Info

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

No preference

IPv4 VPC CIDR block Info

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

20.0.0.0/16

IPv4 subnet CIDR block

20.0.0.0/24

256 IPs

▼ Tags - optional

Key

Value - optional

Q Name X

Q Private_subnet_VPC-B X

Remove

Add new tag

You can add 49 more tags.

Remove

Add new subnet

Cancel

Create subnet

3. Private Route Table Hazırlama

- Route tables → Create route table
- Name: PrivateRT
- VPC: VPC-B

☰

VPC > Route tables > Create route table

🔍 ⌂ 🗨

Create route table Info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

PrivateRT

VPC

The VPC to use for this route table.

vpc-02c87f1940a921715 (VPC-B)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X

Q PrivateRT X

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

- Route table oluşturduktan sonra, **Subnet associations** → **Private-Subnet-B** seçip kaydedin.

(Not: Burada IGW eklemiyoruz çünkü subnet'in internete çıkışı olmayacak.)

VPC > Route tables > rtb-Ode8defd4b0fd3b3f

Route table rtb-Ode8defd4b0fd3b3f | PrivateRT was created successfully.

rtb-Ode8defd4b0fd3b3f / PrivateRT

Details Info

Route table ID: rtb-Ode8defd4b0fd3b3f

VPC: vpc-0fd4a1583c3b26623 | VPC-B

Main: No

Owner ID: 565164711667

Explicit subnet associations: -

Edge associations: -

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
20.0.0.0/24	local	Active	No	Create Route Table

VPC > Route tables > rtb-095e460441cfa6189 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/1)

Filter subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Private_subnet_VPC-B	subnet-Oa7be9f34bcc7fc2e	20.0.0.0/24	-	Main (rtb-02101ec92835c047d)

Selected subnets

subnet-Oa7be9f34bcc7fc2e / Private_subnet_VPC-B

Cancel Save associations

4. Private EC2 Oluşturma

Şimdi private subnet içinde internet erişimi olmayan bir EC2 başlatacağız.

- **EC2** → **Launch instances**
- **Name:** VPC-B_EC2
- **AMI:** Amazon Linux 2023 (kernel-6.1)

Name and tags [Info](#)

Name

VPC-B_EC2

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents

My AMIs

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-00ca32bbc84273381 (64-bit (x86), uefi-preferred) / ami-0aa7db6294d00216f (64-bit (Arm), uefi-preferred)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250818.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-00ca32bbc84273381	2025-08-13	ec2-user

- Instance type: **t2.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- Key pair: Daha önce oluşturduğunuz **ec2_ssh_key.pem** dosyasını seçin.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ec2_ssh_key

[Create new key pair](#)

- Network settings → Edit:
 - VPC: VPC-B
 - Subnet: varsayılan (Private_subnet_VPC-B)
 - Auto-assign public IP: Disable
 - Firewall (SG): Create a new security group
 - Name: Private_EC2_SG
 - Description: Security group for private EC2
 - Rule: SSH → Source: Anywhere (lab için)

Network settings

VPC - required

vpc-02c87f1940a921715 (VPC-B)

Subnet

subnet-0a7be9f34bcc7fc2e Private_subnet_VPC-B

Auto-assign public IP

Disable

Firewall (security groups)

Create security group

Security group name - required

Private_EC2_SG

Description - required

Security group for private EC2

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0)

Type	Protocol	Port range	Source type	Source	Description - optional
ssh	TCP	22	Anywhere	0.0.0.0/0	e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your Instance. We recommend setting security group rules to allow access from known IP addresses only.

Add security group rule

Summary

Number of Instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.8.2...read more

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch Instance

Preview code

- Launch instance düğmesine bas ve → running olana kadar bekle.

Instances (2)

Last updated less than a minute ago

Connect

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type
VPC-B_EC2	i-03885e059e3ab59c3	Running	t2.micro
VPC-A_EC2	i-0267704e3cd7d281a	Running	t2.micro

✓ Bu instance yalnızca **Private IP** alacak. IPv4 Private IP adresini not edin (örneğin: 20.0.1.25).

i-03885e059e3ab59c3 (VPC-B_EC2)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-03885e059e3ab59c3

IPv6 address

—

Hostname type

IP name: ip-20-0-0-37.ec2.internal

Answer private resource DNS name

—

Auto-assigned IP address

—

Public IPv4 address

—

Instance state

Running

Private IP DNS name (IPv4 only)

ip-20-0-0-37.ec2.internal

Instance type

t2.micro

VPC ID

vpc-02c87f1940a921715 (VPC-B)

Private IPv4 addresses

20.0.0.37

Public DNS

—

Elastic IP addresses

—

AWS Compute Optimizer finding

User: arn:aws:iam::538594437814:user/cloud_user is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: * with an explicit deny in a service control policy

Retry

Bölüm 3: İki VPC Arası Bağlantıyı Test Et (Peering Öncesi)

Şimdi elimizde:

- **VPC-A_EC2 (Public bastion)** → Public IP'si var, dışarıdan erişilebilir.
- **VPC-B_EC2 (Private)** → Sadece private IP'ye sahip, internete çıkışı yok.

Amacımız, bastion üzerinden private EC2'ya **özel IP ile SSH** yapabilmek. Ama henüz peering olmadığı için bu bağlantı gerçekleşmeyecek.

1. Bastion'a Bağlan

Önce bilgisayarımızdan bastion sunucuya (VPC-A_EC2) SSH ile bağlanalım:

```
chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem ec2-user@<VPC-A_EC2_PUBLIC_IP>
```

👉 Bu komutla bastion'a giriş yapıyoruz.

2. Key Dosyasını Bastion'a Kopyala

Private EC2'ya da aynı key ile bağlanacağımız için, **.pem** dosyasını localimizden bastion'a göndermeliyiz:

```
scp -i ec2_ssh_key.pem ec2_ssh_key.pem ec2-user@3.89.19.3:/home/ec2-user
```

Bastion'a ssh ile giriş yaptıktan sonra dosya izinlerini sıkılaştırın:

```
ssh -i ec2_ssh_key.pem ec2-user@3.89.19.3
```

```
base) Downloads 👉 ssh -i ec2_ssh_key.pem ec2-user@3.89.19.3
,      #_
~\     ####_      Amazon Linux 2023
~~  \_#####\
~~      \###|
~~      \#/  ___  https://aws.amazon.com/linux/amazon-linux-2023
~~      V~'  '->
~~~
~~~.  _  /
~~~  /  /
~~~ /m/'

ec2-user@ip-10-0-1-168 ~]$ ls -l
total 4
-r----- 1 ec2-user ec2-user 1674 Sep 11 15:26 ec2_ssh_key.pem
ec2-user@ip-10-0-1-168 ~]$
```

```
chmod 400 ec2_ssh_key.pem
```

3. Private EC2'ya Bağlanmayı Dene

Artık bastion üzerinden private EC2'nun **private IP'si** ile SSH denemesi yapalım (örneğin **20.0.1.25**):

```
ssh -i ec2_ssh_key.pem ec2-user@20.0.1.37
```

```

, #_
~\_ #####_ Amazon Linux 2023
~~ \_#####\
~~ \###|
~~ \#/ ____ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~._. _/
_/_/_/
_/_m/'
ec2-user@ip-10-0-1-168 ~]$ ls -l
total 4
-r----- 1 ec2-user ec2-user 1674 Sep 11 15:26 ec2_ssh_key.pem
ec2-user@ip-10-0-1-168 ~]$ chmod 400 ec2_ssh_key.pem
ec2-user@ip-10-0-1-168 ~]$ ssh ec2-user@20.0.0.37 -i ec2_ssh_key.pem

ssh: connect to host 20.0.0.37 port 22: Connection timed out

```

✅ Bu noktada bağlantı **başarısız olacaktır**. Çünkü iki VPC arasında henüz bir iletişim yolu (peering) yok.

Bölüm 4: VPC Peering Oluşturma ve Route Table Güncelleme

Şimdi sıra geldi iki VPC'yi konuşTURacak köprüyü kurmaya: **VPC Peering Connection**.

- AWS Console'da **VPC** → **Peering Connections** → **Create Peering Connection** yoluna gidin.
- **Peering connection name tag:** VPC-A_to_VPC-B
- **VPC ID (Requester):** VPC-A
- **VPC ID (Acceptor):** VPC-B
- **Region:** Aynı region seçili olacak (ör. us-east-1).
- **Account:** Varsayılan bırakabilirsiniz.

Create peering connection

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them privately. [Info](#)

Peering connection settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
VPC-A_to_VPC-B

Select a local VPC to peer with

VPC ID (Requester)
vpc-0021fa599947cef4d (VPC-A)

VPC CIDRs for vpc-0021fa599947cef4d (VPC-A)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☒ This Region (us-east-1)
☐ Another Region

VPC ID (Acceptor)
vpc-02c87f1940a921715 (VPC-B)

VPC CIDRs for vpc-02c87f1940a921715 (VPC-B)

CIDR	Status	Status reason
20.0.0.0/16	Associated	-

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Q Name X

Value - optional
Q VPC-A_to_VPC-B X Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create peering connection](#)

- **Create Peering Connection** butonuna basın.

Peering connections (1) [Info](#)

[Find peering connections by attribute or tag](#)

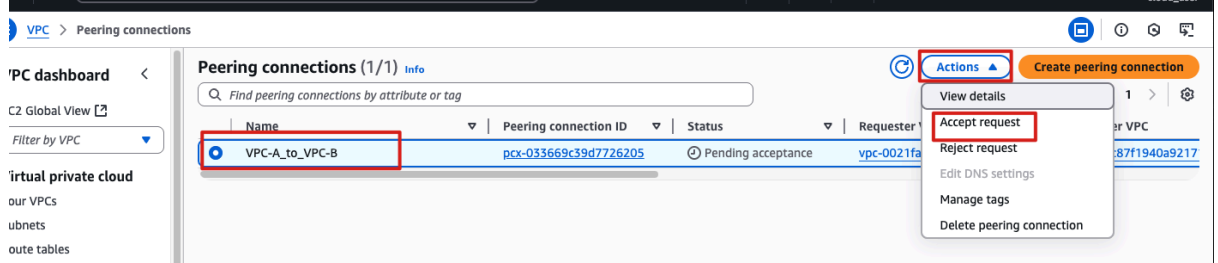
Name	Peering connection ID	Status	Requester VPC	Acceptor VPC
VPC-A_to_VPC-B	pcx-033669c39d7726205	Pending acceptance	vpc-0021fa599947cef4d / VPC-A	vpc-02c87f1940a921715

[Create peering connection](#)

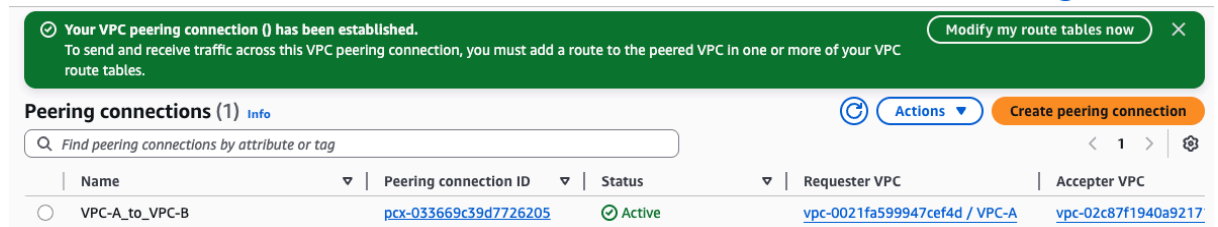
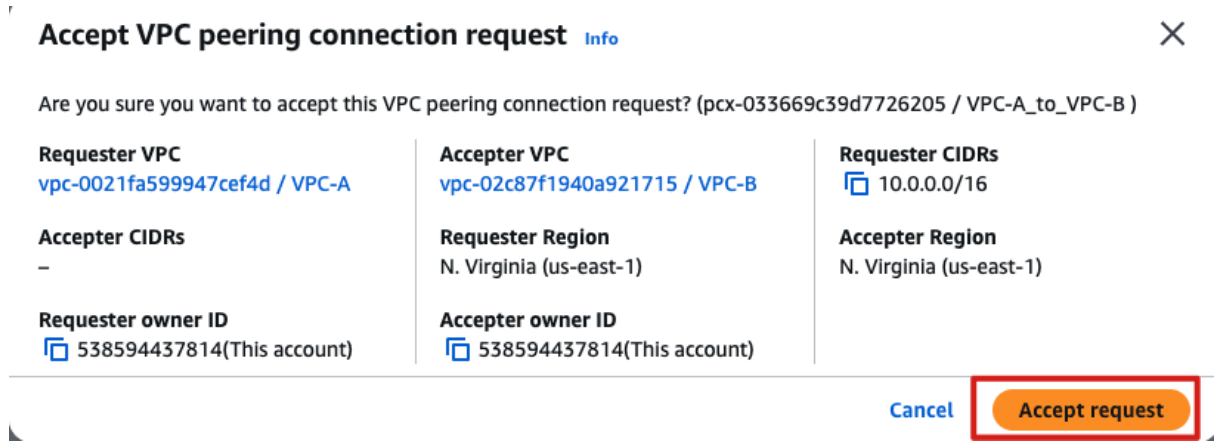
- ♦ Durum ilk etapta **Pending Acceptance** olacaktır.

2. Peering İsteğini Kabul Etme

- VPC → Peering Connections menüsünden oluşturduğunuz bağlantıyı seçin.
- Actions → Accept request tıklayın.
- Artık peering'in durumu **Active** olmalı.



Durum **Active** olmalı (görmüyorsan sayfayı yenile).



VPC-A dan VPC-B ye Peering **aktif** olsa da trafik ancak route ekleyince akar.

3. Route Table Güncellemeleri

Peering bağlantısı aktif olsa da, trafiğin akabilmesi için her iki VPC'nin route tablolarını güncellememiz gerekir.

Route tables→PublicRT-A→edit routes

☰ VPC > Route tables > rtb-0d02c60a36bf898e8 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.0.0.0/16	local	Active	No	CreateRouteTable	
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove
20.0.0.0/16	igw-0b891a26604fa8bcc	-	No	CreateRoute	Remove
	Peering Connection	-	No	CreateRoute	Remove
	pcx-033669c39d7726205	-	No	CreateRoute	Remove

Add route

Cancel Preview **Save changes**

PublicRT-A (VPC-A):

- **Destination:** 20.0.0.0/16
- **Target:** Peering connection (VPC-A_to_VPC-B)

Route tables→PrivateRT→edit routes

☰ VPC > Route tables > rtb-095e460441cfa6189 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
20.0.0.0/16	local	Active	No	CreateRouteTable	
10.0.0.0/16	Peering Connection	-	No	CreateRoute	Remove
	pcx-033669c39d7726205	-	No	CreateRoute	Remove

Add route

Cancel Preview **Save changes**

PrivateRT-B (VPC-B):

- **Destination:** 10.0.0.0/16
- **Target:** Peering connection (VPC-A_to_VPC-B)

🔑 Bu sayede iki VPC arasındaki trafik karşılıklı yönlendirilecektir.

✓ Artık VPC-A'daki bastion üzerinden VPC-B'deki private EC2'ya SSH yapabilirsiniz.

```
ssh ec2-user@20.0.0.37 -i ec2_ssh_key.pem
```

```
[ec2-user@ip-10-0-1-168 ~]$ ssh ec2-user@20.0.0.37 -i ec2_ssh_key.pem
The authenticity of host '20.0.0.37 (20.0.0.37)' can't be established.
ED25519 key fingerprint is SHA256:vhjk8hi2HRzwlh4rxGHLC2M9Cdk9Cte6HMBqUgDztwE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.0.0.37' (ED25519) to the list of known hosts.
,      #_
~\    #####_      Amazon Linux 2023
~~   \#####\
~~    \###|
~~     \#/  ____  https://aws.amazon.com/linux/amazon-linux-2023
~~      V~'  '->
~~~~
~~_.._  _/
  _/_/_/
    _/m/'
[ec2-user@ip-20-0-0-37 ~]$
```


BÖLÜM 5: Kaynakları Temizleme (Cleanup)

Lab ortamlarında yaptığımız her kurulum, AWS üzerinde **gerçek kaynaklar** oluşturur ve bunlar dakikalar içinde fark etmeden maliyet yazmaya başlayabilir. Özellikle **EC2 instance** ve **VPC Peering Connection** gibi kaynaklar, uzun süre açık kalırsa gereksiz masraf çıkarır.

Bu nedenle lab bittikten sonra şu adımları takip ederek ortamı temizleyelim:

1. **EC2 Instance'ları Durdur ve Sil**
 - **VPC-A_EC2** (public bastion)
 - **VPC-B_EC2** (private sunucu)
2. **Subnet'leri Sil**
 - **Public-Subnet-A**
 - **Private-Subnet-B**
3. **Route Tablolarını Sil**
 - **PublicRT-A**
 - **PrivateRT**
4. **Internet Gateway'i Sil**
 - **IGW-A**
5. **VPC Peering Connection'ı Sil**
 - **VPC-A_to_VPC-B**
6. **VPC'leri Sil**
 - **VPC-A**
 - **VPC-B**

Sonuç

Bu lab'da öğrendiklerimizi özetleyelim:

- İki VPC arasında güvenli iletişim kurmak için **VPC Peering** oluşturduk.
- **Public Bastion** → **Private EC2** üzerinden SSH bağlantısını test ettik.
- Peering'in çalışabilmesi için **route tablolarını çift taraflı güncelledik**.
- Lab sonunda kaynakları temizleyerek maliyetleri sıfırladık.

VPC Peering; basit yapısıyla küçük/orta ölçekli senaryolar için idealdir. Ancak çok sayıda VPC'yi birbirine bağlamak gerektiğinde **Transit Gateway** gibi daha merkezi çözümler tercih edilebilir.